

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Tech. Examination (EC/ME/CL)

December 2011

CL 101.01 Fundamentals of Civil Engineering

Date: 13.12.2011, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Enlist principles of building planning. [02]
(b) Define Civil Engineering. Explain role of civil engineer. [05]
- Q - 2 (a) Explain Aspect & Privacy in detail. [04]
(b) Give the functions of Door, Wall, Plinth and Foundation [04]
(c) Write a note on BOT Projects for highway. [06]

OR

- Q - 2 (a) Define Traffic signals & enlist advantages of the same. [02]
(b) Write advantages and disadvantages of water ways. [04]
(c) Explain in detail orientation of building and selection of site. [04]
(d) List out different types of loads & explain any two in detail. [04]
- Q - 3 (a) Give classification of foundation. [02]
(b) Draw a neat sketch of 20 cm thick wall foundation showing all dimensions. [02]
(c) Differentiate between Load bearing structure and Framed structure. [04]
(d) Enlist the different construction material. Write requirements of good bricks. [06]

OR

- Q - 3 (a) Draw the detailed plan for line diagram shown in **figure 1**. Also give the dimensions and position of doors and windows. The width of tread is 20 cm. The plinth height is 60 cm and floor height is 3.0 m. Take scale as 1cm = 0.5 m [08]
(b) What is cement? List out uses of cement. [03]
(c) Write advantages and disadvantages of timber. [03]

SECTION - II

- Q - 4 (a) Differentiate between plan and map. [02]
(b) The magnetic bearing of a line AB is S 28° 30' E. Calculate the true bearing if the declination is 7° 30' W. [03]

P.T.O.

- (c) An area of 49 cm^2 of a map represents an area of 2401 km^2 on field. Find scale and R.F. of the map. [02]

Q - 5 (a) Explain reciprocal ranging. [04]

- (b) The offset from boundary of an agricultural plot are taken as below at the interval of 20 m. Calculate the area between the base line and boundary by Simpson's rule and trapezoidal rule. The offsets are 0.24, 0.89, 1.45, 2.56, 5.67, 8.76, 7.88, 4.56, 3.66, 2.5 and 1.35. [06]

- (c) Write requirements of water. [04]

OR

Q - 5 (a) The following readings are taken on continuously falling ground with staff of 4.0 m. They are 0.400, 0.765, 1.270, 2.560, 3.220, 3.950, 0.390, 1.690, 3.500, 0.800, 1.920, 2.450, and 3.980. The instrument was shifted after 6th and 9th readings. Enter the reading in the page of level book and calculate the RLs of all the points by RISE & FALL method. If the first reading was taken on benchmark of 100.000 m. [07]

- (b) Enlist uses of surveying [03]

- (c) Enlist various water losses. Briefly explain them. [04]

Q - 6 (a) Enlist different types of chains and tapes. [02]

- (b) The observed bearings are mentioned in table below. Find out included angles and [06]

LINE	FB	BB
AB	$20^\circ 30'$	$200^\circ 00'$
BC	$110^\circ 00'$	$290^\circ 30'$
CD	$195^\circ 00'$	$15^\circ 00'$
DA	$286^\circ 30'$	$106^\circ 00'$

also correct the angles if needed.

- (c) Explain with neat sketch types of aquifer. [06]

OR

Q - 6 (a) Define perpendicular and oblique offsets. [02]

- (b) A chain of 30 m was used to measure a rectangular field. Length of longer side was measured to be 110 m and chain was found to be 2 cm too long. Further the work was continued and shorter side of rectangle was measured to be 80 m and the chain was 4 cm too long. Find the area of field in square meter and in hectare. [06]

- (c) Discuss classification of canal. [06]

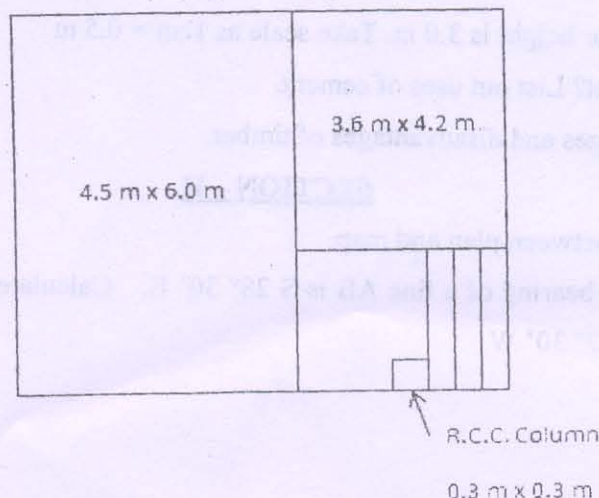


Figure: 1